Muhammad Farhan

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GitHub | LinkedIn

CAREER OBJECTIVE

Results-driven Computer Vision and AI Engineer with hands-on experience building and deploying CV/ML systems in production environments. Passionate about edge AI, deep learning model optimization, and real-time computer vision applications. Seeking to contribute technical expertise in object detection, image recognition, and embedded AI to drive innovation in fleet safety and intelligent transportation systems.

EXPERIENCE

AI Engineer *Dec 2025 – Present*

AI Data House | Islamabad, Pakistan

- Designing and deploying custom computer vision models across multiple providers (OpenAI Vision, Azure AI Vision, AWS Rekognition) and custom-trained architectures, tailored to diverse client requirements.
- Built a hybrid CV pipeline for electrical diagram analysis using OpenAI Vision; extracted component labels and their relationships with 85% accuracy, combining OCR and vision model outputs for structured data extraction.
- Developed and deployed RAG-based AI systems integrating vector search with LLMs; deployed models and APIs on Azure (Azure AI Services, Azure Container Apps) and AWS (EC2, S3) for production-grade inference.
- Delivered an OpenClaw-based AI automation agent (client project) enabling task execution across system, apps, and online services using Anthropic and OpenAI models; ran locally with full data privacy on client hardware.
- Applied deep learning techniques (CNNs, transformers, transfer learning) and machine learning models for real-world visual and NLP tasks; fine-tuned pre-trained models on domain-specific datasets.
- Collaborated with cross-functional teams to deliver end-to-end AI solutions — from dataset preparation and model training to cloud deployment and client integration.

AI Engineer *Jul 2025 – Dec 2025*

NTIS PAK | Islamabad, Pakistan

- Developed, deployed, and optimized AI models for automated translation services using Python, TensorFlow, Keras, and ElevenLabs.
- Led a team in implementing NLP and predictive analytics solutions to replace manual interpretation with AI-driven systems.
- Automated translation workflows, improving service delivery speed, accuracy, and scalability for UK-based clients.

AI Engineer *Mar 2025 – Jul 2025*

Amal Axis | Islamabad, Pakistan

- Developed and maintained AI applications with scalable, reusable code and managed cloud deployments.
- Implemented and optimized ML/DL models; collaborated with cross-functional teams on product delivery.
- Troubleshot performance bottlenecks and stayed current with Python frameworks and ML research trends.

Freelance AI & Full Stack Developer *Feb 2023 – Present*

Remote – Pakistan |

- NextNRGie Platform: Architected full-stack SaaS for client lifecycle management with FastAPI backend, MySQL, CI/CD via GitHub Actions and Docker, and React/Material-UI frontend deployed at nextnrgie.fr.
- Delivered end-to-end AI and full-stack solutions including model training, backend APIs, and responsive frontends for multiple clients.

EDUCATION

Bachelor of Science in Computer Science (BSCS) *Oct 2021 – Jun 2025*
Bahria University Islamabad | Islamabad, Pakistan
CGPA: 3.54 / 4.00 | Thesis: Cartoon Sketch Colorization (AI-based GAN system)
Relevant Courses: Data Structures & Algorithms, Design & Analysis of Algorithms, OOP, Operating Systems, Computer Networks, Digital Image Processing, Introduction to Data Science, Web Engineering, Database Management Systems, Parallel & Distributed Systems, Information Security

PROJECTS

Cartoon Sketch Colorization (Final Year Project) — *PyTorch, Pix2Pix GAN, Stable Diffusion, FastAPI, React, Firebase*

- Built AI-based full-stack platform colorizing black-and-white cartoon sketches using Pix2Pix GAN (TensorFlow/Keras) and generating images from text prompts via Stable Diffusion.
- Designed FastAPI + SQLAlchemy + JWT backend with React.js + Three.js + Framer Motion frontend for interactive 3D UI and smooth animations.
- Integrated Firebase Google OAuth for secure user login and persistent chat-based interaction with saved output history.

Facial Emotion Recognition — *CNN, TensorFlow, OpenCV, FER2013*

- Developed CNN-based ML model for real-time human emotion detection from facial images, classifying 7 emotions with attention mechanism on FER2013 dataset.
- Implemented full data preprocessing pipeline (resizing, grayscale conversion, normalization) and evaluated performance through accuracy/loss visualization.

NextNRGie – Full-Stack SaaS Platform — *FastAPI, MySQL, React, Docker, CI/CD, JWT*

- Architected and deployed contract/estimate/invoice management SaaS with secure JWT authentication, automated PDF generation, Alembic migrations, and multilingual responsive frontend.

Other Notable Projects

- Brain Tumor Classifier: CNN-based MRI classifier (VGG16/ResNet) achieving 92% accuracy on 3,000+ scans.
- Healthcare NLP Chatbot: BERT-powered chatbot for medical queries with context awareness.
- E-commerce Web App: MERN-stack site with user auth, product CRUD, and Stripe integration.
- Patient Management App (Flutter): Flutter app with CSV import/export, SQLite storage, and patient search.

TECHNICAL SKILLS

Computer Vision & ML: Object Detection, Image Segmentation, CNNs, GANs, Transformers, Transfer Learning, Model Compression, Edge AI Deployment
Frameworks & Libraries: PyTorch, TensorFlow, Keras, OpenCV, FastAPI, Flask, Django, React, Next.js, Hugging Face, Stable Diffusion
Programming Languages: Python, JavaScript, C++, Dart, HTML5, CSS3, R
MLOps & Cloud: Docker, CI/CD (GitHub Actions), Azure (AI Services, Container Apps), AWS (EC2, S3), Firebase, Google Colab, RAG Pipelines
Tools & Databases: Git, GitHub, MySQL, PostgreSQL, MongoDB, SQLite, Android Studio, VS Code

HONORS & AWARDS

- Team Lead – NASCON'24 Speed Programming Competition, FAST University.
- Oracle Generative AI Professional Certification (Aug 2025).
- Google Data Analytics Professional Certificate – Coursera (2025).
- Secured A grade in Final Year Project (Cartoon Sketch Colorization AI Platform).

CERTIFICATIONS

- Oracle Generative AI Professional – Oracle (Aug 2025)
- Google Data Analytics Professional Certificate – Coursera / Google (2025)